

**(14 marks) Choose the correct answer

Q1) One of the following systems is linear

- a) $\begin{cases} x_1 - 2x_2 = 3 \\ 2x_1 + (x_2)^2 = 4 \end{cases}$ b) $\begin{cases} x_1 - 2x_2 = \sqrt{2} \\ 2x_1 + (\ln 7)x_2 = \frac{3}{5} \end{cases}$ c) $\begin{cases} x_1 = 3x_2 - 7 \\ 2x_1 - \frac{1}{x_2} = \frac{1}{3} \end{cases}$ d) $\begin{cases} x_1 - 2x_2 = 4 \\ 3x_1 + 5x_1x_2 = 0 \end{cases}$

Q2) If A and B are 3×3 matrices with $\det A = -3$ and $\det B = 8$, then $\det(3B)'(2A)^{-1} =$

- a) $-\frac{64}{81}$ b) -9 c) $-\frac{16}{9}$ d) $-\frac{1}{9}$

Q3) If A and B are 2×2 matrices with $\operatorname{tr} A = 5$, $\operatorname{tr} B = -3$ and $\operatorname{tr} AB = 2$, then $\operatorname{tr}(3BA)' =$

- a) 54 b) -45 c) -135 d) 6

Q4) If A is 5×3 matrix and $AX = YA = A$, then

- a) $X = I_3, Y = I_3$ b) $X = I_3, Y = I_5$ c) $X = I_5, Y = I_3$ d) $X = I_5, Y = I_5$

Q5) One of the following expressions makes sense for any vectors u, v , and w in $\mathbb{R}^n$

- a) $\|u, v\|$ b) $\|u\| \cdot \|v\|$ c) $u \cdot (v \cdot w)$ d) $u \cdot v - 3\|u\|$

Q6) If $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = 2$, then $\det \begin{bmatrix} 2a & c \\ -6b & -3d \end{bmatrix} =$

- a) -72 b) 12 c) -12 d) 72

Q7) If A is $n \times n$ matrix with $A^2 = I - 4A$, then $A^{-1} =$

- a) $A - 4I$ b) $A + 4I$ c) $4A$ d) $-4A$

** (8 marks) Fill in the blanks

Q9) If $A = \begin{bmatrix} 3 & 5 \\ 3 & d \end{bmatrix}$, $C(A) = \begin{bmatrix} 1 & -2 \\ C_{12} & C_{22} \end{bmatrix}$, then $\det A = \dots -7 \dots$
 (Where $C(A)$ is the matrix of cofactors) 4

Q10) If the linear system $\begin{cases} x + 2y = 1 \\ 2x + (a^2 - 5)y = a + 5 \end{cases}$ has No solution, then $a = \dots 4 \dots$

Q11) The solution of the augmented matrix $\begin{bmatrix} 1 & 2 & 3 & 3 & 3 \\ 0 & 0 & 1 & -1 & 1 \end{bmatrix}$ is

$(x_1, x_2, x_3, x_4) = \dots$

Q12) Given the linear system $\begin{cases} ax + by = 1 \\ cx + dy = 2 \end{cases}$, with $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = 2$, $\det \begin{bmatrix} 2 & b \\ 4 & d \end{bmatrix} = 8$, $\det \begin{bmatrix} a & -1 \\ c & -2 \end{bmatrix} = 6$,

then $x = \dots 1 \dots$, $y = \dots 1 \dots$

** (4 marks) Answer the following questions as True or False

Q13) For any $n \times n$ matrix A we have $A'A$ is always a symmetric matrix. True

Q14) $\det(A - B) = \det A - \det B$ false

Q15) If A is $n \times n$ matrix with $\det A = 2$, then the solution of the homogeneous system $AX = 0$ is Only the trivial solution. false

Q16) The sum of two invertible matrices is invertible. True

Q17) (4 marks) Solve the following linear system in detail using Gauss Jordan elimination

$$x_1 + x_2 + 2x_3 = 8$$

$$x_1 + 2x_2 - 3x_3 = -1$$

$$x_1 + 2x_2 + x_3 = 7$$

Unique
Solution

$$\begin{bmatrix} 1 & 1 & 2 & 8 \\ 1 & 2 & -3 & -1 \\ 1 & 2 & 1 & 7 \end{bmatrix}$$



$$\begin{bmatrix} 1 \end{bmatrix}$$

